Position & Evaluation Sub-controller

Agent Design & Operating Specification

A persistent Sub-controller within the Market Development Controller that converts decision-ready research evidence and qualified market signals into disciplined candidate, hypothesis, economic, confidence, comparison, and Market Position assessments. It may evaluate, compare, request targeted additional research, coordinate bounded evaluation work, and recommend Market Position transitions; the Market Manager Agent retains authoritative write ownership of the canonical Market Position.

Field	Value
Loop / Domain	Market Development Loop - Position & Evaluation control domain
Parent Controller	Market Development Controller
Agent type	Persistent Sub-controller / specialist control domain
Version	0.4
Status	Architecture-aligned working specification
Document type	Agent Design & Operating Specification
Architecture basis	Hipstraw Agent Architecture at Scale v2.0; Hipstraw Product & System Design Corpus v0.4; 17 Hipstraw Manager Dimensional Models v0.2; 18 Hipstraw Language, Grammar, Interpretation & Compilation Specification v0.4; current parent/peer Agent Design & Operating Specifications v0.4; Hipstraw Engineering & Validation Pack v0.1.
Canonical Position owner	Market Manager Agent

Specification boundary: this document defines the Position & Evaluation Sub-controller as an intelligent class and persistent operating process. It covers responsibility, decision rights, contracts, lifecycle, class behavior, state, evidence, evaluation semantics, failure, scale, observability, and production readiness. It intentionally does not prescribe database schemas, queue/event-bus products, model vendors, prompt text, concrete API payloads, deployment topology, or low-level scoring algorithms; those belong to Technical Architecture and implementation specifications.

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1. Purpose, Role & Responsibilities

1.1 Operational problem

Hipstraw continuously receives research findings, economic evidence, contradictions, campaign-derived market signals, changes in confidence, and new candidate opportunities. These inputs do not become Market Position truth automatically. They require a persistent evaluation process that can interpret evidence, compare alternatives, identify unresolved uncertainty, and produce traceable recommendations without bypassing the Market Manager Agent.

Without a dedicated Position & Evaluation process, the Market Manager Agent would either need to perform detailed evaluation itself or rely on worker outputs that may be incomplete, inconsistent, stale, or mutually contradictory. The result would be weak separation of authority, unstable Position updates, duplicated analysis, and poor auditability.

1.2 Purpose

Purpose
Convert evidence and material market signals into reliable, traceable Position assessments and recommendations, while preserving the Market Manager Agent as the authoritative owner of the canonical Market Position.

1.3 Central question

Central question
Given the available evidence, what should Hipstraw conclude about a candidate or existing Market Position, how confident should it be, what remains unknown, and what should happen next?

1.4 Expected outcomes

- Decision-ready assessment of a candidate, hypothesis, or existing Market Position.
- Explicit evidence support, contradictions, unknowns, economic interpretation, and confidence.
- Clear recommendation: maintain, strengthen, weaken, investigate further, validate, defer, challenge, recommend transition, recommend retirement, or escalate.
- Bounded research/validation requests when evidence is insufficient.
- Durable evaluation history and lineage suitable for later review, learning, and audit.

1.5 Architectural position in the Market Development Loop

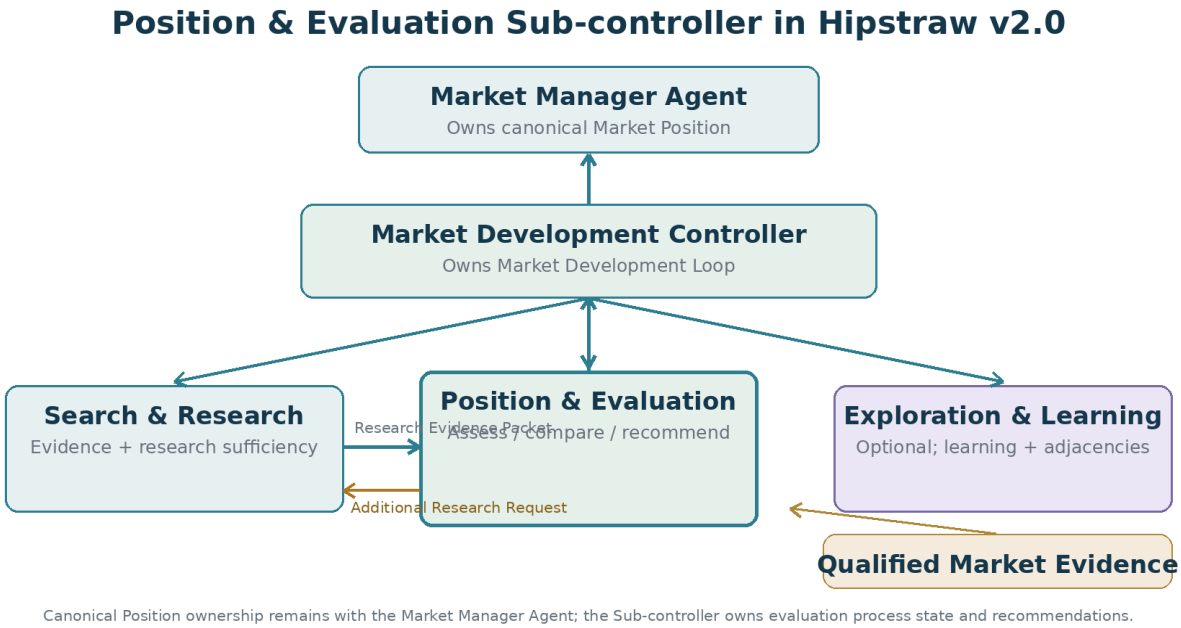


Figure 1. Position & Evaluation Sub-controller in Hipstraw Agent Architecture at Scale v2.0

The Market Manager Agent owns canonical Market Position. The Market Development Controller owns the Market Development Loop and delegates detailed evaluation control to the Position & Evaluation Sub-controller. Search & Research supplies Research Evidence Packets; Position & Evaluation interprets and compares that evidence, assesses economics/confidence, and returns structured Position recommendations. Exploration & Learning may supply bounded learning/adjacency hypotheses when enabled. Qualified Campaign Market Evidence may enter evaluation only when it is consequential enough to cross the loop boundary.

1.6 Core responsibilities

The Sub-controller owns the persistent evaluation-control domain inside the Market Development Controller: evaluation intake and prioritization; evidence-readiness gating; candidate/hypothesis and economic assessment; comparative evaluation; confidence and contradiction assessment; evaluation-specific Worker coordination; targeted Additional Research Requests; Position/transition recommendations; and durable evaluation state, lineage, and completion status.

1.7 Responsibilities explicitly not owned

It does not own canonical Market Position write authority, portfolio-level resource allocation, Seed Graph semantics or traversal, primary evidence acquisition, campaign strategy/execution, material policy exceptions, or human-reserved high-consequence decisions.

1.8 Design invariants

Evidence is not Position. Evaluation output is not canonical belief. Position & Evaluation may assess and recommend; Market Manager Agent establishes canonical Market Position through Review. Intent and bounded authority flow downward; evidence, uncertainty, recommendations, and escalation flow upward. Workers scale bounded analysis, not strategic authority. Higher-consequence recommendations require stronger evidence and assurance.

2. Scope, Boundaries & Decision Rights

2.1 Responsibilities owned

Control responsibility	What the sub-controller owns
Evaluation intake	Receive, validate, prioritize, schedule and track evaluation requests/events.
Candidate / hypothesis assessment	Evaluate evidence for support, contradiction, materiality, market potential and unresolved uncertainty.
Economic evaluation	Interpret available economic evidence and compare expected opportunity under approved rubrics.
Candidate comparison	Compare multiple candidates or position alternatives on common decision dimensions.
Confidence assessment	Assess and update confidence based on evidence quality, consistency, coverage and freshness.
Research/validation commissioning	Request additional Search & Research work or bounded worker analysis when evidence is insufficient.
Recommendation formation	Prepare Position recommendations and transition rationale for Market Development Controller synthesis / Market Manager Agent Review.
Evaluation state	Maintain queue, dependencies, worker assignments, unresolved issues, decision history, wait/wake state and completion status.
Evaluation assurance	Apply rubrics, independent evaluation, challenger analysis and stronger review when consequence is high.

2.2 Responsibilities not owned

- Authoritative write ownership of the canonical Market Position.
- Final Market Position activation, retirement, or other major Position transition.
- Program-level market priority or resource allocation.
- Primary ownership of search, graph traversal, evidence acquisition, or source discovery.
- Primary ownership of Campaign Position or Campaign Development & Execution Loop operation.
- Unbounded self-generated agendas outside the Market Manager Agent / Program mandate.
- Policy exceptions or high-consequence actions reserved for the Market Manager Agent or human authority.

2.3 Decision-rights model

Authority rule
The Position & Evaluation Sub-controller may assess, compare, request more evidence, and recommend. The Market Manager Agent decides the authoritative Market Position. The sub-controller may update only its own provisional/evaluation state and shared assessment artifacts within delegated authority.

2.4 Boundary with Search & Research Sub-controller

Search & Research Sub-controller	Position & Evaluation Sub-controller
Determines what evidence should be acquired and how.	Determines what available evidence implies for a candidate/hypothesis/Position.
Operates search space/frontier and research execution.	Operates evaluation queue, comparison, confidence, and recommendation process.
Returns Research Evidence Packets.	Returns research/validation requests when evidence is inadequate.
Owns research sufficiency for the research task.	Owns evaluation sufficiency for making a Position recommendation.

2.5 Boundary with Exploration & Learning Sub-controller

When enabled, Exploration & Learning Sub-controller turns consequential outcomes, anomalies, and repeated signals into bounded learning and adjacency hypotheses. Position & Evaluation may assess their implications, but does not create exploration programs or treat hypotheses as market truth. Formal evidence acquisition remains with Search & Research; canonical Market Position remains with the Market Manager Agent.

2.6 Boundary with Market Development Controller

The Market Development Controller is the parent authority and owns the Market Development Loop. It supplies evaluation mandate, priority, resource envelope, decision need, applicable constraints, and the current market-development context. Position & Evaluation owns detailed evaluation control and returns structured assessments, comparisons, Position recommendations, Additional Research Requests, unresolved uncertainty, and escalations.

2.7 Boundary with Market Manager Agent

The Market Manager Agent owns canonical Market Position, major Position transitions, market-side strategic priority, and resource direction. Position & Evaluation must never silently write canonical Market Position; it prepares traceable recommendations with evidence, economics, confidence, contradictions, unknowns, and required assurance for Manager Review.

2.8 Human approval boundary

Human approval remains required where program governance reserves high-consequence actions, policy exceptions, material external commitments, or other decisions beyond delegated agent authority. The Sub-controller must escalate rather than manufacture authority or certainty.

2.9 Boundary with Worker Agents

Evaluation Workers perform bounded economics, comparison, independent evaluation, challenger analysis, evidence validation, sensitivity analysis, and related specialist work. They return typed results; they do not own the persistent evaluation queue, recommendation process, or canonical Market Position.

3. Inputs, Outputs, Managed Objects & Interaction Contracts

3.1 Inputs

Input	Source	Purpose
Evaluation Request	Market Development Controller / scheduled review / Sub-controller wake event	Defines the object, decision need, scope, priority and deadline.
Research Evidence Packet	Search & Research Sub-controller	Primary evidence basis for evaluation.
Current Market Position snapshot	Shared canonical state / Market Manager Agent	Provides current interpreted market state and version.
Market hypotheses / candidate set	Shared market state	Defines propositions or alternatives to assess.
Economic assessments / estimates	Research workers or shared evidence state	Supports opportunity-value interpretation and candidate comparison.
Contradictions / unknowns	Shared evidence state	Identifies unresolved uncertainty requiring explicit treatment.
Material Campaign signal	Filtered Campaign Development & Execution Loop coupling	Adds consequential behavioral evidence from active market engagement.
Policies, rubrics, goals, guardrails	Program / governance state	Constrains judgment, authority, evidence thresholds and escalation.

3.2 Managed objects

Managed object	Definition
Evaluation Request	Durable unit of evaluation demand with objective, scope, priority, dependencies, authority, state and completion criteria.
Candidate Assessment	Structured assessment of one candidate market/opportunity against approved dimensions.
Hypothesis Assessment	Support/contradiction/uncertainty assessment for a market hypothesis.
Comparison Set	Two or more candidates or alternatives evaluated under the same rubric and evidence cut.
Economic Assessment	Interpreted economic potential, assumptions, ranges, sensitivities and confidence.
Confidence Assessment	Explicit confidence state and rationale based on evidence quality and consistency.

Managed object	Definition
Research / Validation Requirement	Bounded request for evidence needed to resolve a material gap or contradiction.
Position Recommendation	Proposed Position change with evidence, confidence, economics, implications and rationale.
Evaluation Record	Durable ledger entry capturing inputs, decisions, workers, rubrics, outputs and lineage.

3.3 Outputs

Output	Primary destination	Purpose
Position Assessment	Market Development Controller / shared assessment state	Summarizes what the evidence currently supports and does not support.
Candidate Comparison	Market Development Controller / Market Development Loop	Makes alternatives comparable without hiding uncertainty.
Confidence Update	Shared assessment state	Records evaluation-level confidence changes and reasons.
Research / Validation Request	Search & Research Sub-controller	Requests missing evidence or contradiction resolution.
Position Transition Recommendation	Market Development Controller	Recommends maintain, strengthen, weaken, defer, challenge, transition, retire, or escalate for parent synthesis and Manager Review.
Evaluation Escalation	Market Development Controller / higher authority	Surfaces unresolved, high-consequence, policy-bounded, or authority-bounded issues.
Evaluation Record	Ledger / audit state	Preserves provenance and decision history.

3.4 Interaction contract requirements

Every handoff must identify the governed scope, requested decision, referenced Position/evidence versions, required output type, authority and budget envelope, evidence/assurance standard, completion condition, and escalation route. The Sub-controller should reject or clarify malformed requests rather than infer unlimited scope or authority.

3.5 Acceptance & completion contracts

An evaluation case is accepted only when subject/scope, decision need, evidence context, authority, and applicable constraints are resolvable. It is complete when a typed recommendation/assessment has been

persisted and routed, a targeted research request has been issued, the case is explicitly waiting/deferred, the mandate is withdrawn/superseded, or the unresolved issue has been escalated.

4. Operating Model, Lifecycle & Prioritization

4.1 Core operating loop

Operating loop
Receive -> Contextualize -> Evaluate -> Compare -> Assess economics/confidence -> Decide sufficiency -> Recommend / Request Research / Escalate -> Persist -> Return / Wait / Stop

The sub-controller is a persistent logical actor, not a continuously running model process. It may remain asleep while retaining identity, subscriptions, queue ownership and durable state; cognition occurs through Agent Runs when a wake condition requires evaluation work.

4.2 Intake, activation & wake conditions

Wake condition	Typical action
New evaluation request	Validate, enqueue and prioritize.
ResearchEvidenceReady	Resume or start evaluation using new evidence.
MaterialCampaignSignal	Assess whether observed behavior materially changes a market belief.
EvidenceStale	Re-open evaluation or request refreshed evidence.
PositionReviewDue	Perform scheduled evaluation against current Position and evidence.
ContradictionDetected	Assess materiality and commission validation/challenge if needed.
Threshold crossing	Re-evaluate when confidence, economics, momentum or other governed threshold changes.
Deadline / human request	Prioritize required evaluation or escalation.
Dependency resolved	Resume blocked evaluation from durable state.

4.3 Queue, prioritization & next-action selection

The sub-controller maintains a durable evaluation queue. Queue priority should reflect Market Manager Agent / Market Development Controller priority, decision consequence, evidence readiness, urgency, expected economic importance, freshness, dependencies, cost of delay, and the value of resolving uncertainty. Exact weights remain policy/configuration rather than class identity.

4.4 Lifecycle states

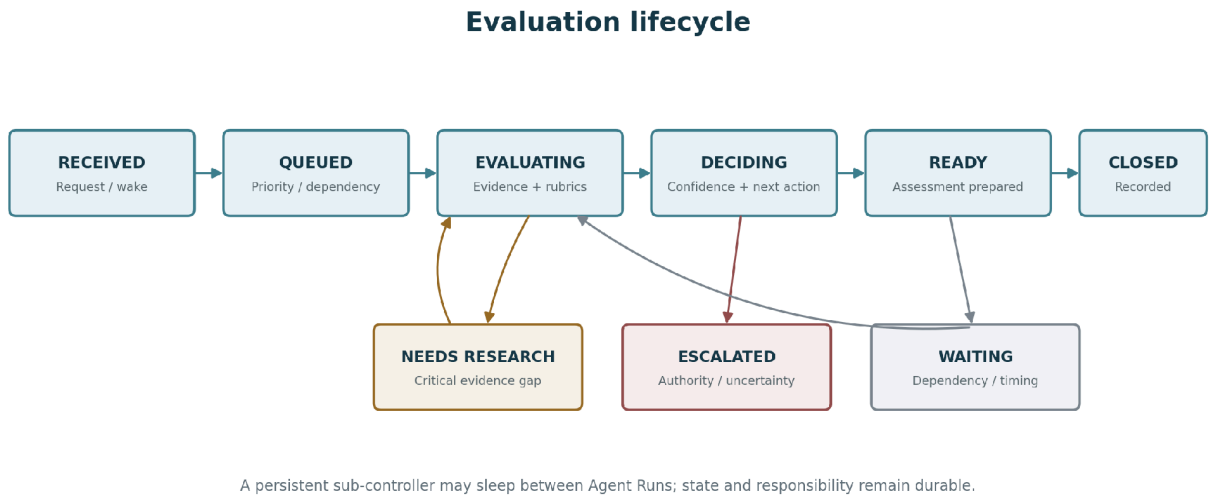


Figure 2. Evaluation lifecycle and side states

4.5 Concurrency and version control

- Multiple candidates may be evaluated concurrently when work is independent.
- One candidate may use parallel specialist or independent-evaluation workers, but the sub-controller maintains one coherent evaluation context and comparison basis.
- Position recommendations must reference the Market Position version and evidence cut they evaluated. If the canonical Market Position changes before recommendation acceptance, the sub-controller must revalidate or mark the recommendation stale.
- Conflicting parallel evaluations are treated as useful evidence and reconciled through rubrics, additional validation, challenger analysis, or escalation rather than silently averaged.

4.6 Completion conditions

- Evidence is sufficient to produce a bounded recommendation at the required consequence level.
- Insufficient evidence has been explicitly identified and a research/validation request issued.
- The evaluation is deferred or waiting on a defined dependency or timing condition.
- The issue exceeds delegated authority or cannot be resolved and has been escalated.
- The evaluation is no longer required because the candidate/request was withdrawn or superseded.

4.7 Evaluation planning

For each selected case, determine evaluation objective, applicable Rubric(s), evidence-readiness requirement, economic analysis depth, comparison set, assurance level, Worker needs, dependencies, budget, stop condition, and output contract before executing material analytical work.

4.8 Wait, sleep, wake & resume

The Sub-controller is logically persistent but computationally intermittent. It may WAIT/SLEEP when no executable action exists. On wake, it reloads the durable Evaluation Case, revalidates referenced Position/evidence versions and constraints, and resumes from the last valid control point.

4.9 Completion & stopping

Stop when evidence is sufficient for the required bounded recommendation, the case is blocked and a targeted research request or escalation has been issued, expected decision value becomes immaterial, the mandate is withdrawn/superseded, or policy/budget/time requires termination.

4.10 Concurrent evaluations & backpressure

Independent cases may run concurrently, but the Sub-controller must bound active work, preserve coherent per-case versions, and apply backpressure when queue depth, cost, latency, Worker capacity, or downstream review capacity would otherwise create uncontrolled execution.

5. Decision & Evaluation Model

This section defines the recurring decisions that make the sub-controller a persistent cognitive process rather than a passive synthesis step. Exact scoring formulas and rubric weights remain configurable, but the decision classes and authority boundaries are part of the class design.

#	Control point	Primary question	Typical outcome
1	Evaluation eligibility	Is the request sufficiently scoped, authorized, current and decision-relevant?	Accept / merge / defer / reject / escalate
2	Evidence sufficiency	Is the evidence adequate for the requested evaluation and consequence level?	Evaluate / request research / wait
3	Hypothesis assessment	Does evidence support, weaken, contradict, or leave the hypothesis unresolved?	Supported / weakened / contradicted / unresolved
4	Candidate assessment	How attractive/material is the candidate under the approved rubric?	Assessment with evidence, economics, confidence and gaps
5	Economic assessment	What economic opportunity is supported, under what assumptions and uncertainty?	Range / assumptions / confidence / sensitivities
6	Candidate comparison	How do alternatives compare on common dimensions without hiding evidence differences?	Relative assessment / trade-offs / unresolved differences
7	Confidence assessment	How confident should Hipstraw be in the evaluation?	Confidence state + rationale + limiting factors
8	Unknown / contradiction treatment	Could unresolved uncertainty materially change the conclusion?	Research / validate / challenge / tolerate / escalate
9	Position recommendation	What Position change, if any, should be recommended?	Maintain / strengthen / weaken / defer / transition / retire / no recommendation

#	Control point	Primary question	Typical outcome
10	Completion / escalation	Can the sub-controller responsibly complete within authority?	Complete / wait / request research / escalate

5.1 Evidence-readiness gate & standard control-point contract

For every material decision Inputs -> Decision -> Applicable rubric/policy -> Skills -> Tools -> Evidence basis -> Output -> State transition -> Authority -> Escalation condition

5.2 Evidence interpretation, economics & candidate evaluation

Separate observed evidence from Worker/model interpretation and from Sub-controller synthesis. Evaluate material supporting and weakening evidence, assumptions, freshness, provenance, contradictions, opportunity mechanism, value pool, cost to capture, time to value, durability, uncertainty/sensitivity, problem strength, product/scenario fit, buyer accessibility, timing, adoption feasibility, and competitive/alternative pressure under the active Rubric.

5.3 Comparative evaluation & confidence

Compare candidates using a common active Rubric and evidence policy unless a documented exception applies. Keep attractiveness and confidence separate; surface near-ties and trade-offs; use independent/challenger evaluation when consequence or closeness warrants it. Confidence is a meta-assessment based on evidence quality, coverage, independence, freshness, consistency, contradictions, assumption sensitivity, and consequence - not a model self-rating.

5.4 Research-sufficiency & Position implication decision

Determine whether the evidence supports a bounded Position implication now, requires targeted additional evidence, should remain unresolved/deferred, or exceeds delegated authority. Absence of evidence must not be treated as evidence of absence unless the applicable search/evidence policy supports that inference.

5.5 Transition recommendation

Recommendations may maintain, strengthen, weaken, defer, investigate further, challenge, deprioritize, recommend transition, recommend retirement, or escalate. They are recommendations to the Market Development Controller / Market Manager Agent, never silent canonical writes.

5.6 Position recommendation states

Recommendation	Meaning
Maintain	Current Position remains supported at the present level.
Strengthen	Evidence materially increases support, confidence, or opportunity potential.
Weaken	Evidence materially reduces support, confidence, or opportunity potential.
Investigate Further	A material unknown could change the evaluation and is researchable.

Recommendation	Meaning
Validate	A consequential claim or economic assumption requires stronger independent support.
Defer	Decision should wait because timing, dependency, or evidence readiness is inadequate.
Challenge	Current Position should be adversarially tested before a consequential transition.
Recommend Transition	Evidence supports movement to a materially different Position state.
Recommend Retirement	Evidence indicates the Position should no longer consume meaningful attention/resources.
Escalate / No recommendation	Authority, uncertainty, contradiction, or policy prevents a responsible recommendation.

5.7 Illustrative Kozmo evaluation

Example: Kozmo is exploring economic opportunities across invoices, delivery, future spend, and the supplier portfolio. Search & Research returns evidence that mid-market technology companies with complex professional-services spend show high invoice complexity and material upcoming commitments, but buyer urgency is uncertain. The Position & Evaluation Sub-controller may assess the candidate as economically promising with moderate confidence, request targeted buyer/urgency validation, and withhold a stronger Market Position recommendation until that uncertainty is resolved. The Market Manager Agent remains responsible for the authoritative canonical Market Position decision through Review.

6. Class Design, Capabilities, Skills, Tools & Worker Delegation

6.1 Class identity, hierarchy & cardinality

Class attribute	Working design
Class name	PositionEvaluationSubController
Base class	HipstrawController / PositionEvaluationSubController under MarketDevelopmentController
Persistence	Persistent identity and durable state; cognition occurs through episodic Agent Runs.
Default scope	One persistent instance per independently governed Market Development scope / Program unless an explicit architecture decision defines another cardinality.
Cardinality rule	Scale Workers with workload first; add another persistent sub-controller only when a workstream has independent state, queue, decisions, resource control and consequence.

Class attribute	Working design
Canonical Position ownership	None. Market Manager Agent owns Market Position.
Primary resource boundary	Delegated evaluation budget, worker concurrency, model/tool usage and review capacity.

6.2 Parent Controller, persistence & Agent Run lifecycle

Parent: Market Development Controller. Default persistence: one durable Position & Evaluation Sub-controller instance per independently governed Market Development scope unless an explicit architecture decision defines another cardinality. Identity, queue/process state, subscriptions, and decision history persist across intermittent Agent Runs; cognition activates on wake conditions and returns to wait/sleep when no useful authorized work remains.

6.3 Native Sub-controller capabilities

Innate capability	Required behavior
Evaluation Intake	Receive, validate, deduplicate and classify evaluation requests/events.
Queue & Priority Management	Maintain durable queue, priorities, dependencies, schedules and wait states.
Evaluation Process Management	Create and progress evaluation instances through defined states and completion criteria.
Task / Worker Orchestration	Issue Work Orders, coordinate specialist/independent/challenger workers, and evaluate results.
State Management	Maintain evaluation context, checkpoints, unresolved issues, decision history and Position/evidence versions.
Confidence Management	Track confidence changes and limiting factors across an evaluation.
Completion Management	Determine when a recommendation is ready, incomplete, waiting, stale or escalated.
Backpressure / Concurrency Control	Bound simultaneous work according to priority, budget and review capacity.
Escalation	Recognize authority, evidence, policy or consequence limits and package escalation.
Recovery / Resume	Retry, replace workers, reconcile state conflicts, resume after wake, and preserve evaluation lineage.
Audit / Ledger Recording	Persist material decisions, evidence references, rubrics, methods and outcomes.

6.4 Required Skills

Skill	Purpose
Evidence Synthesis	Synthesize multi-source evidence without erasing contradictions or unknowns.
Hypothesis Analysis	Assess support, weakening, contradiction and alternative explanations.
Candidate Evaluation	Assess a candidate against market, audience, problem, category, offer and opportunity dimensions.
Economic Evaluation	Interpret opportunity potential, assumptions, ranges, sensitivity and uncertainty.
Candidate Comparison	Compare alternatives on a common evidence/rubric basis.
Confidence Assessment	Estimate confidence using evidence quality, coverage, freshness, independence and consistency.
Contradiction Analysis	Diagnose conflicting evidence and determine validation, challenge, tolerance or escalation.
Position Assessment	Translate evaluated evidence into a bounded Market Position recommendation.
Challenger / Independent Evaluation	Test strong conclusions for blind spots, disconfirming evidence and alternative interpretations.
Temporal / Trajectory Reasoning	Consider freshness, momentum, expected trajectory and whether evidence has decayed.

6.5 Tool classes

Tool class	Purpose
Shared evidence store	Read evidence, claims, contradictions, freshness and provenance.
Market Position read interface	Read current canonical Position and version; no authoritative write ownership.
Evaluation / rubric service	Apply versioned rubrics, thresholds and comparison dimensions.
Model gateway	Invoke approved reasoning, synthesis, extraction or independent-evaluation models.
Analytics / computation tools	Perform bounded quantitative comparisons, sensitivity or economic calculations.
Research record / document retrieval	Inspect Research Evidence Packets, prior assessments and supporting artifacts.
Ledger / history	Read and write evaluation lineage, decisions and recommendation history.

Tool class	Purpose
State / queue service	Maintain persistent process state, dependencies, schedules and Agent Run checkpoints.
Worker runtime	Instantiate bounded specialist, independent-evaluation or challenger workers.
Policy / governance service	Check authority, evidence thresholds, review requirements and constraints.

6.6 Conceptual methods / behavioral interfaces

Method	Responsibility
receiveEvaluationRequest	Validate and persist a new evaluation request/event.
prioritizeEvaluation	Set/revise priority using manager intent, consequence, urgency and evidence readiness.
evaluateCandidate	Assess one candidate under approved rubric and evidence cut.
evaluateHypothesis	Assess support/contradiction/unknowns for a hypothesis.
assessEconomics	Interpret opportunity potential and uncertainty.
compareCandidates	Produce a common-basis comparison and trade-offs.
assessConfidence	Update confidence and limiting factors.
requestResearch	Issue bounded Research / Validation Requirement upstream.
commissionWorker	Create a Work Order for bounded specialist evaluation.
recommendPositionTransition	Produce a Position Recommendation for Market Development Controller synthesis / Market Manager Agent Review.
escalate	Package unresolved or authority-bounded issue.
wait	Persist state and wake condition.
resume	Restore state after wake and continue from a valid checkpoint.
closeEvaluation	Record completion, output, ledger and dependencies.

6.7 Capability -> Skill -> Tool routing

Routing follows Agent Class -> native capability -> Skill -> Tool. The Sub-controller selects a Skill/Tool/Worker combination based on the evaluation question, consequence, evidence type, required independence, latency/cost, policy, and assurance level; Tools supply mechanism, Skills supply institutional competence, and native capabilities define how the class operates.

6.8 Evaluation Worker Swarm & Work Orders

The Sub-controller may instantiate bounded Evaluation Workers for economic analysis, comparison, evidence validation, independent evaluation, challenger analysis, or other specialist work. Each Work Order must define objective, context, inputs, scope, required typed result, evidence standard, permitted Skills/Tools, authority, budget, stop condition, and escalation rule. Worker failure must be replaceable without loss of durable Evaluation Case state.

7. State, Evidence, Rubrics, Economics, Confidence & Assurance

7.1 Durable evaluation state / Evaluation Case

State domain	Examples
Queue state	Request ID, priority, state, deadline, dependency, wake condition.
Evaluation context	Candidate/hypothesis/Position version, decision need, scope and applicable rubrics.
Work state	Worker assignments, attempts, results, pending validation, comparison set.
Issue state	Contradictions, unknowns, stale evidence, unresolved economics, authority questions.
Resource state	Delegated budget, concurrency, model/tool usage, remaining review capacity.
Decision state	Assessments, confidence changes, next action, recommendation history, escalation history.
Recovery state	Last stable checkpoint, interrupted operation, retry/fallback status.

Minimum Evaluation Case state includes: evaluation_case_id; governed subject/candidate; objective and requested decision; current status; referenced Market Position version; evidence version/cut; active Rubric version; economic assumptions; comparison set; contradictions; unknowns; confidence; dependencies; Worker assignments; authority/budget envelope; decision history; recommendation; completion status; and next wake/action condition.

7.2 Shared state read or updated

- Canonical Market Position: read-only for this sub-controller; authoritative writes belong to Market Manager Agent.
- Shared evidence: claims, sources, provenance, freshness, contradictions, unknowns and confidence metadata.
- Candidate markets / hypotheses / economic assessments: shared durable domain records.
- Campaign-derived Market signals: filtered, material cross-loop inputs.
- Ledger / outcomes: provenance of prior beliefs and later reality-testing.

7.3 Rubrics & evaluation dimensions

Rubrics structure judgment and should remain versioned, testable, explainable and distinct from Skills. A minimum working catalog for this sub-controller may include:

Rubric	What it judges
Market Hypothesis Rubric	Strength and materiality of the underlying market belief.
Candidate Evaluation Rubric	Fit, problem materiality, buyer relevance, market potential and evidence quality.
Economic Opportunity Rubric	Economic impact, plausibility, assumptions, sensitivity and confidence.
Evidence Quality Rubric	Authority, relevance, freshness, independence, specificity, coverage and contradiction.
Validation Rubric	Whether a claim/hypothesis is sufficiently supported, contradicted or unresolved.
Position Transition Rubric	Whether evidence justifies maintain/strengthen/weaken/defer/transition/retire recommendation.

7.4 Economics, confidence & uncertainty

Confidence should not be a model self-rating. It should be derived from evidence quality, coverage, freshness, independence, consistency, unresolved contradictions, sensitivity of economic assumptions, and the consequence level of the recommendation. The representation may be numeric, banded, or multi-dimensional as long as it remains calibrated, versioned, and accompanied by limiting factors.

7.5 Assurance levels & decision history

Consequence / uncertainty	Minimum assurance pattern
Low	Single evaluation path with deterministic validation of required fields and evidence lineage.
Moderate	Specialist evaluation plus evidence-quality checks and explicit contradiction handling.
High	Independent evaluation, challenger analysis, stronger corroboration, sensitivity analysis and Market Development Controller synthesis / Market Manager Agent Review.
Very high / policy-bounded	Escalation to Market Manager Agent and human authority where governance requires it.

Provenance principle
Position = what Hipstraw currently believes. Ledger = how Hipstraw arrived at that belief. Outcome = what actually happened. The sub-controller contributes to the Ledger; it does not own canonical Position truth.

8. Events, Signals & Handoffs

8.1 Handoff model & routing

From -> To	Contract	Required content
Search & Research Sub-controller -> Position & Evaluation Sub-controller	Research Evidence Packet	Objective, candidate/hypothesis, claims, sources, contradictions, unknowns, confidence, freshness, research history and limitations.
Position & Evaluation Sub-controller -> Search & Research Sub-controller	Research / Validation Requirement	Decision blocked, exact evidence gap, why it matters, required evidence standard, priority, budget/limit, stop condition.
Position & Evaluation Sub-controller -> Market Development Controller	Position Assessment / Position Recommendation	Current Position/version, assessment, evidence basis, economics, confidence, unknowns, implications, proposed transition, required assurance, and rationale.
Market Development Controller -> Position & Evaluation Sub-controller	Evaluation Mandate / decision request	Object to evaluate, requested decision, priority, consequence, resource envelope, deadline, policy/rubric context, authority, and required assurance.
Campaign Development & Execution Loop -> Market Development Loop	Material Market Signal	Pattern/materiality, confidence, novelty, evidence, urgency and cross-Position relevance.
Position & Evaluation Sub-controller -> Worker	Evaluation Work Order	Objective, input evidence, scope, rubric, output schema, tools, authority, budget, stop condition and escalation rule.

8.2 Events / signals consumed & wake triggers

- EvaluationRequested
- ResearchEvidenceReady
- MaterialCampaignSignal
- EvidenceChanged
- EvidenceStale
- ContradictionDetected
- PositionReviewDue
- ThresholdCrossed
- DependencyResolved
- ManagerDecisionRecorded

8.3 Events / signals emitted

- EvaluationStarted

- EvaluationBlocked
- MoreResearchRequired
- ValidationRequired
- CandidateAssessmentReady
- PositionAssessmentReady
- PositionTransitionRecommended
- ConfidenceChanged
- EvaluationEscalated
- EvaluationClosed

8.4 Materiality, signal filtering & re-evaluation

The sub-controller should not consume every low-level Campaign event. Cross-loop information should be filtered for materiality, confidence, pattern strength, novelty, urgency, evidence quality, and relevance to Market Position. A single negative response may remain in Campaign state; a repeated objection pattern across a meaningful cohort may become a Market Development signal.

8.5 Search & Research handoff

When evidence is insufficient, emit an Additional Research Request that states the blocked decision, material gap, why it matters, required evidence, priority, budget/scope constraints, stop condition, and expected Research Evidence Packet. Search & Research owns how to acquire that evidence.

8.6 Market Development Controller / Market Manager handoff

Return Position Assessment, Economic Assessment, comparison, confidence, contradictions/unknowns, recommendation, required assurance, and resource/learning implications. The parent Controller synthesizes loop-level implications; the Market Manager Agent establishes canonical Market Position through Review.

8.7 Exploration & Learning / Campaign-derived evidence

Qualified Campaign Market Evidence and learning hypotheses may enter evaluation when they are material and scoped. Low-level campaign events remain outside market-side canonical belief. Exploration & Learning may propose adjacencies; Position & Evaluation assesses implications only after suitable evidence exists.

9. Failure, Recovery, Scale, Observability & Production Requirements

Condition	Required behavior
Insufficient evidence	Request targeted research/validation or return unresolved assessment; do not manufacture recommendation.
Contradictory evidence	Preserve competing evidence; assess source/evidence quality; commission verification/challenge; lower confidence or escalate.
Low confidence	Identify limiting factors; determine whether confidence can be improved economically; wait, research, or escalate.
Worker disagreement	Treat divergence as evidence; apply rubric, independent evaluation or challenger rather than simple averaging.
Stale evidence	Mark affected conclusions stale; request refresh or reduce confidence before recommendation.

Condition	Required behavior
Tool / model failure	Retry where transient, switch approved path/model/tool, use partial completion, or escalate.
State conflict / stale Position version	Stop recommendation finalization; reload canonical Position; reconcile or re-run affected evaluation.
Concurrent evaluation conflict	Use versioned evaluation records; synthesize or escalate materially divergent conclusions.
Budget / resource limit	Defer, narrow scope, request authorization, or return bounded incomplete assessment.
Authority violation	Block action and escalate; sub-controller cannot write canonical Market Position.
Unresolved high-consequence uncertainty	Escalate to Market Manager Agent / human authority with evidence and competing interpretations.

9.1 Failure taxonomy

Default recovery order
Validate failure -> Retry transient step -> Replace Worker / model -> Use alternative approved tool or analysis -> Narrow scope -> Use partial evidence with lower confidence -> Wait on dependency -> Escalate

9.2 Escalation package

Escalation contract
Issue -> Decision blocked -> Evidence -> Competing interpretations -> Confidence / consequence -> Work already attempted -> Resource / policy implication -> Recommended higher-authority decision

9.3 Retry, fallback & recovery

Worker failure or model failure should not collapse the persistent evaluation process. Replace execution capacity and preserve the durable evaluation state, evidence references, decisions, and lineage needed to resume.

9.4 Scale, concurrency & backpressure

Support many queued Evaluation Cases without treating them as simultaneously active. Bound active concurrency by resource envelope, consequence, Worker capacity, model/tool limits, cost, and downstream review capacity. Scale Worker execution before multiplying strategic authority or persistent Sub-controller instances.

9.5 Idempotency, stale input & version conflict

Repeated processing of the same case/evidence version must not create duplicate recommendations or transitions. Recommendations must reference the evaluated Market Position/evidence/Rubric versions. If material inputs change before acceptance, mark the result stale and revalidate or re-evaluate.

9.6 Observability & auditability

Record case state transitions, queue age, decision latency, evidence readiness, Worker/model/tool usage, retries, cost, confidence, contradiction handling, research requests, recommendation outcome, escalation, failures, and lineage sufficient to reconstruct why a recommendation was made.

9.7 Production requirements & safe degradation

Production behavior requires durable state, resumability, bounded retries, replaceable Worker/tool execution, version conflict protection, policy enforcement outside prompt prose, cost/budget controls, auditable decisions, and the ability to degrade safely to waiting, lower-confidence recommendation, targeted research request, or escalation instead of manufacturing certainty.

10. Scenarios, Evaluation & Production Readiness

10.1 Critical scenario suite

Scenario	Pass condition
Normal candidate evaluation	Evidence is adequate; candidate is assessed, confidence set, recommendation prepared and ledger updated.
Insufficient evidence	Sub-controller identifies material gap and requests targeted Search & Research work rather than forcing conclusion.
Strongly supported hypothesis	Evidence and economics support strengthening; independent check used if consequence warrants.
Contradicted hypothesis	Credible disconfirming evidence weakens/overturns support and is reflected in recommendation.
Two close candidates	Comparison preserves trade-offs, evidence differences and uncertainty instead of declaring an artificial winner.
Economic estimates disagree	Sensitivity and assumption analysis performed; unresolved material disagreement reduces confidence or escalates.
New evidence weakens existing Position	Position version is read; recommendation to weaken/reassess is sent to Market Manager Agent.
Material Campaign signal	Repeated behavioral evidence is evaluated for Market Position implications; low-level noise is filtered out.
Stale evidence	Affected assessment is marked stale and refresh requested before consequential recommendation.
Low-confidence evaluation	Unknowns/limitations are explicit; no unjustified Position transition recommendation.
High-consequence transition	Independent evaluation/challenger and stronger evidence/review are required.
Concurrent evaluations	Versioning and reconciliation prevent conflicting recommendations from corrupting state.
Tool / worker failure	State persists; execution is replaced/retried without restarting the whole evaluation.

Scenario	Pass condition
Sleep -> wake -> resume	No executable work -> persist -> sleep -> wake event -> reload -> resume correctly.
Authority-boundary test	Attempted canonical Position write is blocked and routed to Market Manager Agent.

10.2 Evaluation metrics & quality dimensions

Dimension	Example measures
Decision quality	Expert agreement with assessment/recommendation; reversal/rework rate; consistency across equivalent cases.
Evidence quality	Provenance coverage, freshness, contradiction handling, unknown visibility and source-quality compliance.
Comparison quality	Comparable basis, trade-off clarity, absence of hidden evidence asymmetry.
Confidence calibration	Observed accuracy/outcomes by confidence band; ability to identify limiting factors.
Research request quality	Percentage of requested research that materially resolves decision uncertainty; unnecessary research rate.
Authority compliance	No unauthorized canonical Position writes; correct escalation of high-consequence issues.
Runtime reliability	Queue integrity, resume success, failure recovery, stale-state conflict rate.
Cost / latency	Time and resource use to complete evaluation by consequence level.
Traceability	Recommendation can be reconstructed from Position version, evidence, rubric, workers and decisions.
Downstream usefulness	Market Manager Agent acceptance/rework and whether recommendations improve subsequent learning/outcomes.

10.3 Falsification / design-risk tests

- The Market Manager Agent repeatedly has to redo the evaluation because the sub-controller output is not decision-usable.
- The sub-controller begins to function as the de facto canonical Market Position owner.
- Research requests are broad, repetitive, or do not reduce material uncertainty.
- Confidence remains high despite weak, stale or contradictory evidence.
- Parallel evaluations routinely create inconsistent recommendations that cannot be reconciled.
- The cost/latency of maintaining a persistent sub-controller exceeds the value of specialized evaluation.
- Position & Evaluation repeatedly requires the Market Development Controller or Market Manager Agent to redo detailed evaluation because its outputs are not decision-usable, indicating the control domain is not earning its persistence.

10.4 Production-readiness stages & exit criteria

Readiness	Minimum evidence of readiness
Not Ready	Design exists but lifecycle, authority and decision model are not implemented end to end.
Experimental	Core intake, evaluation, research-request, recommendation and state persistence work in simulation/internal tests.
Controlled	Bounded real programs demonstrate durable state, evidence discipline, confidence handling, authority compliance and recovery.
Validated	Scenario suite and workload tests show stable decision quality, low Market Manager Agent rework, calibrated confidence and traceable recommendations.
Production Ready	Continuous operation meets target governance, scale, recovery, observability, quality, cost and latency requirements.

10.5 Open risks / unresolved design decisions

Thresholds, Rubric weights, confidence representation, consequence tiers, default Worker independence/challenger rules, performance SLOs, and exact human-approval triggers remain configurable governance decisions. They should be validated through the scenario/eval suite rather than frozen prematurely in class semantics.

Appendix A. Decision Rights Matrix

Decision	Position & Evaluation Sub-controller	Market Development Controller / Market Manager Agent	Worker Agents
Prioritize evaluation queue	Decide within delegated priorities	Controller sets/overrides loop priority; Manager sets market-side strategic priority.	No
Evaluate evidence / candidate / hypothesis	Decide	Controller receives assessment; Manager reviews consequential Position implications.	Bounded analysis only
Request more research / validation	Decide within budget/scope	Controller may override/expand scope or resource envelope.	Executes Work Orders

Decision	Position & Evaluation Sub-controller	Market Development Controller / Market Manager Agent	Worker Agents
Commission evaluation workers	Decide within resource limits	Controller sets delegated resource envelope; no task-level scheduling.	N/A
Set evaluation-level confidence	Decide	Controller consumes assessment; Manager establishes Position-level confidence through Review.	Recommend / provide local analysis
Recommend Position transition	Recommend	Controller synthesizes/recommends upward; Manager decides canonical transition.	No
Write canonical Market Position	No	Controller: No. Market Manager Agent: authoritative write through Review.	No
Resolve material cross-loop conflict	Escalate / recommend	Controller coordinates market-loop handling; Manager resolves Position-level implication.	No
Policy exception / high-consequence action	Escalate	Controller escalates; Manager decides or routes to human authority.	No

Appendix B. Evaluation Case & State Transition Reference

From	Trigger / decision	To
RECEIVED	Validated and authorized	QUEUED
QUEUED	Selected for execution	EVALUATING
EVALUATING	Critical evidence gap	NEEDS RESEARCH
NEEDS RESEARCH	ResearchEvidenceReady	EVALUATING

From	Trigger / decision	To
EVALUATING	Dependency / timing	WAITING
WAITING	Wake condition met	EVALUATING or QUEUED
EVALUATING	Authority / unresolved high consequence	ESCALATED
ESCALATED	Manager decision / new mandate	QUEUED / EVALUATING / CLOSED
EVALUATING	Evidence adequate	DECIDING
DECIDING	Recommendation prepared	READY
READY	Output/ledger recorded	CLOSED
READY	Canonical Position version changed	QUEUED / EVALUATING

Appendix C. Capability -> Skill -> Tool Matrix

Capability	Primary skills	Primary tool classes
Evaluation intake / queue	Evidence-gap interpretation; prioritization reasoning	State/queue; policy service; shared Position/evidence read
Candidate / hypothesis evaluation	Candidate Evaluation; Hypothesis Analysis; Evidence Synthesis	Evidence store; rubric service; model gateway
Economic assessment	Economic Evaluation; sensitivity reasoning	Analytics/computation; model gateway; evidence store
Candidate comparison	Candidate Comparison; Confidence Assessment	Rubric service; analytics; shared evidence
Contradiction handling	Contradiction Analysis; Challenger Reasoning	Evidence store; independent workers; model gateway
Confidence management	Confidence Assessment; Evidence Synthesis	Evidence metadata; rubric service; ledger
Research commissioning	Evidence-Gap Analysis; Position Assessment	Interaction contract; worker/search sub-controller interface
Position recommendation	Position Assessment; Temporal/Trajectory Reasoning	Market Position read; ledger; policy/governance
Recovery / resume	State interpretation; bounded replanning	State/queue; worker runtime; observability

Appendix D. Interaction Contract Matrix

D.1 Research / Validation Requirement

Field	Example
Decision blocked	Whether candidate segment should be strengthened to High-Potential.
Evidence gap	Buyer urgency and budget ownership are not sufficiently supported.
Why material	Low urgency could materially reduce near-term opportunity potential.
Required evidence	Recent authoritative/independent evidence of urgency, budget ownership and trigger conditions.
Priority	High - current Position review blocked.
Stop condition	Sufficient evidence to classify urgency and budget ownership, or evidence ceiling reached.
Return format	Updated Research Evidence Packet with supporting/contradicting evidence, confidence and unknowns.

D.2 Position Recommendation

Field	Required content
Position/version	Canonical Market Position ID and version evaluated.
Recommendation	Maintain / strengthen / weaken / defer / transition / retire / no recommendation.
Assessment summary	What changed and why.
Evidence basis	Material supporting and contradicting evidence with lineage.
Economic interpretation	Opportunity potential, assumptions, ranges/sensitivity and confidence.
Confidence	Evaluation confidence and limiting factors.
Unknowns	Material unresolved questions.
Implications	What changes if accepted: learning priority, audience/problem/offer assumptions, research needs or campaign implications.
Authority / review	Required Market Manager Agent/human approval level.

Appendix E. Position Recommendation Example

Illustrative Kozmo example

Candidate: mid-market technology companies with high professional-services supplier spend. Evidence supports high invoice complexity and material future commitments; delivery verification is moderate; buyer urgency is still uncertain. Economic opportunity appears promising but sensitivity to buyer urgency is high. Recommendation: Investigate Further / Validate buyer urgency before strengthening the Market Position. Confidence: moderate. Required next action: targeted buyer/trigger research. Canonical Position change: none until Market Development Controller synthesis / Market Manager Agent Review.

Appendix F. Candidate Comparison Example

Example comparison: Candidate A may show higher estimated economic potential but lower buyer-access evidence and higher uncertainty; Candidate B may show lower potential but stronger evidence, clearer urgency, and lower cost to test. The Sub-controller should preserve dimension-level differences, evidence quality, assumptions, and confidence rather than forcing a single winner from a composite score. A close comparison may legitimately produce a recommendation to run targeted research or bounded campaign validation before advancing either candidate.

Appendix G. Scenario / Eval / Readiness Matrix

Scenario	Primary control points	Expected output
Normal candidate	Eligibility; candidate assessment; economics; confidence; recommendation	Position Assessment + Recommendation
Insufficient evidence	Evidence sufficiency; unknown treatment	Research / Validation Requirement
Contradiction	Evidence sufficiency; contradiction handling; confidence	Validation/challenge + lower confidence or escalation
Close candidates	Candidate comparison; confidence	Trade-off comparison without forced winner
Campaign signal	Eligibility; hypothesis assessment; recommendation	Assessment of Position implication
Stale evidence	Evidence sufficiency; confidence	Refresh request / reduced confidence
High-consequence transition	Assurance; challenger; escalation	Robust recommendation + required review
Concurrent conflict	Version control; comparison; escalation	Reconciled evaluation or escalation
Failure/resume	Recovery; state management	Resume from checkpoint; no lost lineage
Authority boundary	Recommendation vs canonical write	Blocked write + Market Manager Agent handoff

End of specification. Technical Architecture should translate these behavioral contracts into concrete schemas, APIs, eventing, persistence, model/tool routing, concurrency controls and deployment choices without changing the authority, lifecycle or decision semantics defined here.

Canonical v0.4 alignment

Canonical authority: Hipstraw Agent Architecture at Scale v2.0; Hipstraw Product & System Design Corpus v0.4; 17 Hipstraw Manager Dimensional Models v0.2; 18 Hipstraw Language, Grammar, Interpretation & Compilation Specification v0.4; current parent/peer Agent Design & Operating Specifications v0.4; Hipstraw Engineering & Validation Pack v0.1.

Manager dimensional interface

Aspect	Contract
Primary dimensions	M1-M16 assessment/recommendation
Derived support	May analyze implications for M17/M18 but cannot commit them
Write boundary	No M-dimension or Market Position state write

Dimensional boundary
Position & Evaluation is the strongest lower-layer contributor to Market Review, but its outputs remain assessments/recommendations.

Dimension contribution contract

When this Agent produces evidence or an assessment that may affect Manager judgment, the output should identify impacted_dimension_ids, evidence/provenance, proposed implication (if any), confidence/limitations, contradictions/freshness, criticality implication, and transition implication. proposed_state is advisory until the owning Manager Review commits the dimension state.